

Environmentally extended input–output analysis on a city scale – application to Aveiro (Portugal)



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ABSTRACT

This study evaluates direct and indirect environmental impacts associated with the consumption of goods and services by the households living in the municipality of Aveiro (Portugal) using environmentally extended input output (IO) analysis. The environmental impacts addressed are greenhouse gas (GHG) emissions and fossil fuel consumption. A methodology adapted to the specificities of the data available is proposed. This methodology relies on the Portuguese economic IO table, on GHG emission and fossil fuel consumption intensities, and on national household expenditure data downscaled to the urban level. This methodology could also be applied to other municipalities or cities for which data on local household expenditures are missing. This study also identifies the sectors with the largest impacts in order to prioritise strategies for reducing the impacts to move the municipality towards more sustainable consumption.

The estimated total GHG emissions in 2005 amounted to 26 kg CO₂-eq./cap/day and the total fossil fuel consumption was 7.3 koe/cap/day. The sectors that contributed most for both indicators are “land transport and transport by pipelines”, “food, beverages and tobacco”, “construction” and “production, collection and distribution of electricity”. Therefore, improvement measures are proposed for these sectors in order to more effectively decrease the environmental impacts related with GHG emissions and fossil energy consumption of the municipality.

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1. Introduction

There is a growing concern about the environmental impacts of cities. According to the United Nations (UN, 2011), the worldwide urban population was 729 million inhabitants in 1950 (29% of the total population) and went up to 3486 million inhabitants in 2010 (51% of the total population). This is an increase of almost five times in only six decades and it is expected that the urban population will reach 6286 million citizens by 2050. This high increase of population in cities brings along a high increase on demand of products and resources. Hence, high pressures on the environment result from these demands of city dwellers.

A sustainable city implies an urban region for which the inflows of materials and energy and the disposal of wastes do not exceed the capacity of its hinterlands (Kennedy et al., 2007). Urban

metabolism studies are becoming more popular nowadays to track and improve the sustainability of cities (Baynes and Wiedmann, 2012; Zhang, 2013). Urban metabolism is a relatively new field of study introduced in the 1960s by Wolman (1965). It analyses a city describing its functionalities, assessing its environmental impacts, showing the inter-linkages and mutual impacts among urban areas and between urban and rural areas and identifying drivers and possible response measures (Minx et al., 2011).

The environmental assessment of a city or other territory is complex and currently there is no consensual and widely adopted method (Loiseau et al., 2012). A number of holistic methods and tools have been used for this purpose, such as the ecological footprint (Collins et al., 2006; Global Footprint Network, 2013), material flow analysis and accounting (Barles, 2009; Browne et al., 2011; Druckman et al., 2008; Niza et al., 2009), substance flow analysis (Timmermans and Van Holderbeke, 2004; Wu et al., 2012), emergy analysis (Campbell et al., 2004; Pulselli et al., 2007; Zhang et al., 2009), exergy analysis (Sciubba et al., 2008), environmentally extended input–output (IO) analysis (Larsen and Hertwich, 2010; Lenzen and Peters, 2009),

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physical IO tables (Liang and Zhang, 2011), ecological network analysis (Zhang et al., 2010), and hybrid IO Life Cycle Assessment (LCA) (Heinonen et al., 2011; Heinonen and Junnila, 2011).

IO analysis is a macroeconomic technique that uses data on inter-industrial monetary transactions to account for the complex interdependencies of industries in modern economies (Lenzen et al., 2004a). It was introduced by Leontief (1936) and, since then, it has been applied to evaluate several economic, social and environmental issues. An environmentally extended IO analysis links environmental data (e.g. carbon dioxide (CO₂) emissions) for all economic sectors in an economy with monetary fluxes between these sectors (intermediate demand) and allows for an allocation of these environmental burdens to the consumption of product groups (final demand) (Minx et al., 2009). Several studies have applied environmentally extended IO analysis to estimate environmental indicators, such as ecological, carbon and water footprints, at national and supra-national levels (e.g. Andrew and Forgie, 2008; Cruz and Barata, 2008; Hertwich and Peters, 2009; Hubacek et al., 2009; Wiedmann et al., 2010; Zhao et al., 2009). There are only a few studies applying environmentally extended IO analysis to smaller spatial areas such as municipalities or cities, namely Sydney (Lenzen et al., 2004a), Lisbon (Rosado and Ferrão, 2009), Chongqing (Okadera et al., 2006), Zhangye (Wang et al., 2009), Beijing (Wang et al., 2013; Zhang et al., 2011) and five cities in the UK (Minx et al., 2011).

One objective of this study is to provide, for the first time, an overview of the environmental impacts associated with the consumption of goods and services by the households living in the municipality of Aveiro (Portugal), using IO analysis. The environmental impacts addressed are greenhouse gas (GHG) emissions and fossil fuel consumption. Both direct and indirect impacts are considered. Direct impacts arise directly from household activities (e.g. transportation, heating and lighting), whereas indirect impacts result from the production of the goods and services required by the households (e.g. production of fuels, building materials and food). To accomplish this objective, it was necessary to adapt the methodology to the specificities of the available data. These data, which include the IO table, household expenditures, annual GHG emissions and annual fossil fuel consumption, are only available at the national scale. Therefore, a downscaling procedure was developed and proposed in order to estimate municipal GHG emissions and fossil fuel consumption. Another objective consists in identifying the sectors that contribute most to the impacts in order to prioritise strategies for reducing the impacts to move the municipality towards more sustainable consumption.

The municipality of Aveiro is located in the coastal central part of Portugal and has nowadays about 78,450 inhabitants, which corresponds to 0.7% of the total population of the country (INE, 2012). It is composed of 14 civil parish subdivisions with a total area of about 200 km² and an average population density of about 390 inhabitants per km² (INE, 2012). About 70% of the population lives in 5 more urban-like parish subdivisions with population density ranging from 785 to 1385 inhabitants per km² (INE, 2012). The remaining parish subdivisions are rural-like with population density varying between 70 and 400 inhabitants per km² (INE, 2012). Aveiro has a temperate climate with average air temperatures ranging from 6 to 24 °C and an average annual precipitation of 944 mm with rain occurring mainly in autumn and winter (IPMA, 2013).

2. Methodology

2.1. Description of the methodology

In this study, a consumption perspective is adopted, i.e., GHG emissions and fossil fuel consumption are accounted for the

activities required by households living in Aveiro. Due to limitations in data availability, impacts derived from the consumption associated with public services (e.g. street lighting) and waste and wastewater treatment were excluded.

This study is based on an environmentally extended IO analysis that combines the Portuguese economic IO table with environmental data and household expenditure data. All these data are available at the national scale and, therefore, a downscaling procedure was developed and proposed to derive municipal household expenditure data allowing the calculation of GHG emissions and fossil fuel consumption at the municipal level on a daily per capita basis. The overall methodology is schematically illustrated in Fig. 1 and includes the sequential steps described in Sections 2.1.1–2.1.5.

2.1.1. Calculation of the Leontief inverse matrix

First, the matrix A for Portugal was obtained by dividing the national IO table by the gross output of each economic sector (also given in the IO table). The matrix A is the matrix of coefficients that indicate the amount of one economic sector product needed to produce one unit of another economic sector product. Then, the inverse Leontief matrix $(I - A)^{-1}$ (where I is the identity matrix, i.e., with “1” in the diagonal and “0” in all the other fields) was calculated.

According to the conventional IO equation (Eq. (1)), when the Leontief inverse matrix is multiplied by the final consumption vector of a country (Y), the resulting vector is the gross output for that country (X):

$$X = (I - A)^{-1}Y \quad (1)$$

If another consumption vector for spatial scales smaller than the national one (y) is multiplied by the Leontief inverse matrix, the result is a vector of gross output (x) needed to supply that smaller demand. This consumption vector (y) could be defined for a region, a municipality, a city, a household or an individual consumption.

2.1.2. Calculation of the household consumption vector

The second step consisted in determining the household consumption vector for the municipality of Aveiro (y_{Aveiro}) for each economic sector. As data specific for Aveiro about household expenditures are not available, this vector was estimated using average household expenditures for the whole country and applying the downscaling procedures described in Section 2.2.2.

2.1.3. Calculation of the gross output needed to supply household demands

In this step, the gross output of all economic sectors needed to supply Aveiro household demands (x_{Aveiro}) was calculated by multiplying the Leontief inverse matrix (calculated as described in Section 2.1.1) by the household consumption vector (y_{Aveiro} , calculated as described in Section 2.1.2).

2.1.4. Calculation of the environmental intensities

This step consisted in the calculation of GHG emission and fossil fuel consumption intensities at national level (expressed per monetary unit of gross output) of each economic sector based on annual GHG emissions and annual fossil fuel consumption divided by the gross output of each economical sector available in the IO table.

2.1.5. Calculation of the environmental impacts

The environmental impacts (GHG emission and fossil fuel consumption) per capita caused by household demands were obtained by multiplying the gross output of each economic sectors due to household demands (x_{Aveiro} , calculated as described in Section

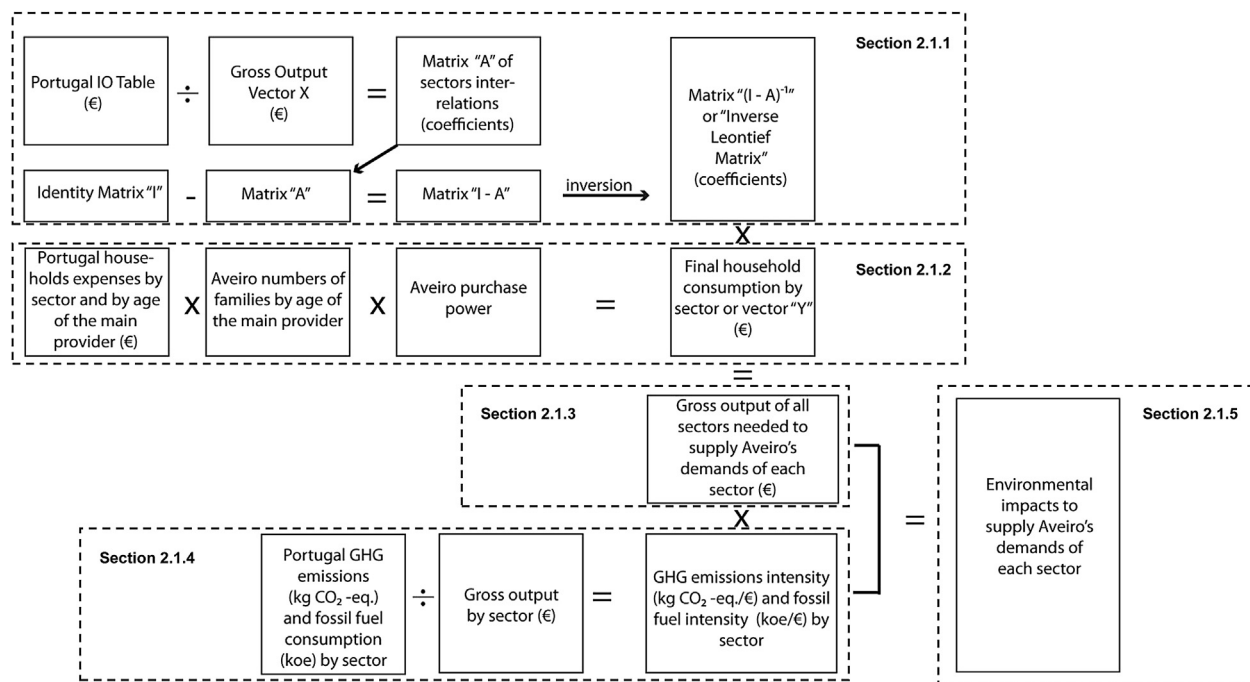


Fig. 1. Synthesis of the methodology used in this study.

2.1.3) by the environmental intensities of each economic sector at national level (calculated as described in Section 2.1.4).

2.2. Data requirements

Most of the data used refer to the year 2005, the only exception being the number of families in Aveiro, which refers to the year 2001 as this type of data is only available in Portugal each period of 10 years (1991, 2001, and 2011) through Population Census inquiries. Year 2001 was selected due to be the year more close to year 2005.

2.2.1. IO table for Portugal

The latest edition of the IO table for Portugal, which refers to 2005 (OECD, 2011), was used in this study. This IO table comprises 48 economic sectors. The Portuguese IO table lacks information for the monetary fluxes of some sectors, namely: "mining and quarrying (energy)", "pharmaceuticals", "non-ferrous metals", "aircraft and spacecraft", "railroad equipment and transport equipment not elsewhere specified" and "manufacture of gas, distribution of gaseous fuels through mains". This lack of information has the drawback that only impacts associated with final consumption are considered for these sectors, while the impacts from intermediate uses are not taken into account.

2.2.2. Household consumption vector of Aveiro

The household consumption vector for the municipality of Aveiro was calculated using the Inquiry of Household Expenditures 2005/2006 (INE, 2008). This inquiry contains data for Portugal on the average annual expenditures per family disaggregated by family type, region and age of the reference person (main income provider). Data disaggregated by family type were not used in this study because the classification of family types in the inquiry is not the same as in the statistical data of the number of families per city (INE, 2002). Since Aveiro is a municipality much more developed than the other municipalities located in the same region, it was decided not to use the regional data as well. Therefore, data on the

household expenditures disaggregated by age of the reference person was considered a better option. These data are provided for a list of 12 sectors subdivided into 47 sub-sectors. Equivalencies were established between these sub-sectors and 22 economic sectors of the IO table (Table A.1 in Appendix). Since these data are provided per family, they needed to be multiplied by the number of families in Aveiro disaggregated by age category of the reference person sourced from the Population Census of 2001 (INE, 2002). Besides, in 2005, the purchase power of Aveiro was 1.34 times higher than the average purchase power in Portugal (INE, 2007) and, therefore, the household expenditures were also multiplied by this factor.

The obtained household consumption vector amounted to a total of 8438 €/cap/year and is presented in Table 1 disaggregated by economic sector of the IO table. Both the money spent for buying a house and paying a rent was assigned to the construction sector, because both are linked to this sector's investments. Ideally, only buying or renting a newly built house should be allocated to the construction sector, but the available data on household expenditures do not make a distinction between money spent on newly built and old houses. Therefore, this is a worst case scenario that placed the construction sector in the first position concerning household annual expenditure with 1758 €/cap/year. The sector of food, beverages and tobacco was in second place with 1510 €/cap/year. Other major expenses were associated with the sectors of hotels and restaurants (920 €/cap/year) and land transport (704 €/cap/year).

2.2.3. Environmental intensities of each economic sector

Average GHG emission and fossil fuel consumption intensities of each economic sector were calculated using the gross output of each sector available in the IO table and the annual national GHG emissions and fossil fuel consumption available in the 2005 National Accounts of the Portuguese Statistics Institute (INE) disaggregated by 81 economic sectors (INE, 2011a,b). Aggregation of some sectors was necessary for the sake of compatibility with the 48 economic sectors of the IO table. Besides, GHG emission and

Table 1

Household final consumption per sector of the IO table in 2005 (€/cap/year).

Economic sector	Aveiro	Portugal
Construction	1758	1332
Food products, beverages and tobacco	1510	1137
Hotels and restaurants	920	692
Land transport; transport via pipelines	704	522
Other community, social and personal services	551	415
Motor vehicles, trailers and semi-trailers	413	301
Textiles, textile products, leather and footwear	351	263
Other manufacturing; recycling (including furniture)	343	260
Pharmaceuticals	289	223
Post and telecommunications	251	188
Production, collection and distribution of electricity	233	176
Health and social work	201	163
Finance and insurance	197	146
Collection, purification and distribution of water	128	97
Education	128	109
Manufacture of gas; distribution of gaseous fuels through mains	93	70
Pulp, paper, paper products, printing and publishing	80	61
Radio, television and communication equipment	63	47
Other electrical machinery and apparatus	77	58
Other machinery and equipment	59	45
Air transport	57	46
Wood and products of wood and cork	32	24
Total	8438	6375

fossil fuel intensities of the economic sectors in the IO table without data on gross outputs (as referred in Section 2.1) and of the economic sectors without data on annual GHG emissions and fossil fuel consumption were assumed to be equal to those of similar sectors as shown in Table A.2 (e.g. pharmaceuticals were considered to have the same intensity of other chemicals). An additional assumption, usual in this type of studies (Lenzen et al., 2004a; Druckman and Jackson, 2009), is that imported goods have the same GHG emission and fossil fuel intensities as those produced in Portugal.

The obtained GHG emission intensity (kg of all GHGs converted into CO₂ equivalent according to their global warming potential – kg CO₂-eq./€) and fossil fuel intensity (kg of oil equivalent – koe/€) are displayed in Table 2 for some selected economic sectors of the IO table.

2.3. Sensitivity analysis

Available data on household expenditure refer to the national scale and they needed to be downscaled to the municipality level, which had introduced some uncertainty. Therefore, a sensitivity analysis was undertaken to evaluate the influence of using different estimates of household expenditure, as follows:

- Scenario A adopts for Aveiro the same consumption pattern of Portuguese households in general (INE, 2008), which amounted to a total of 6375 €/cap/year (Table 1). Therefore, this scenario reports results at the country level;
- Scenario B considers a total expenditure of 8543 €/cap/year obtained by taking into account the per capita national average expenditure (6375 €/cap/year) multiplied by 1.34 (ratio between the purchase power in Aveiro and the national purchase power);
- Scenario C considers a total expenditure of 6297 €/cap/year obtained by applying only partially the downscaling procedure of household expenditures. Data on expenditures disaggregated by age of the reference person were considered but the fact that the purchase power of Aveiro was 1.34 times higher than the national purchase power was disregarded.

Table 2

GHG emission intensity and fossil fuel intensity per monetary unit of gross output of some economic sectors of the IO table for 2005.

Economic sector	GHG emissions intensity (kg CO ₂ -eq./€)	Fossil fuel intensity (koe/€)
Production, collection and distribution of electricity	2.632	0.708
Land transport; transport via pipelines	2.287	0.760
Other non-metallic mineral products	1.990	0.424
Agriculture, hunting, forestry and fishing	1.343	0.058
Pharmaceuticals	0.878	0.201
Wholesale and retail trade; repairs	0.126	0.049
Iron and steel	0.124	0.076
Textiles, textile products, leather and footwear	0.120	0.061
Construction	0.104	0.036
Food products, beverages and tobacco	0.091	0.044
Health and social work	0.060	0.030
Hotels and restaurants	0.044	0.037
Collection, purification and distribution of water	0.022	0.032
Other manufacturing; recycling (including furniture)	0.018	0.012
Education	0.014	0.011
Motor vehicles, trailers and semi-trailers	0.003	0.005

3. Results

Total GHG emissions triggered by household consumption, including direct and indirect emissions, amounted to 26 kg CO₂-eq./cap/day. More than 70% of these emissions arise from the demand of goods and services from four economic sectors: “land transport and transport by pipelines” (22% of the total emissions), “food products, beverages and tobacco” (20% of the total emissions), “construction” (18% of the total emissions), and “production, collection and distribution of electricity” (12% of the total emissions), as shown in Table 3.

The household demand for “land transport and transport by pipelines” was the one that released more GHG emissions over the whole supply chain (5.6 kg CO₂-eq./cap/day). Most of these emissions were directly emitted during household transportation.

GHG emissions derived from the consumption of “food products, beverages and tobacco”, which totalled 5.0 kg CO₂-eq./cap/day, had a much higher dependency on other sectors. About half of these emissions came from the sector of “agriculture, hunting, forestry and fishing”, where food ready to be consumed or raw materials for the food, beverages and tobacco industry are produced. Other significant contributions are associated with the production and supply of the electricity used along the supply chain (~14%) and with the industry itself to produce processed food, beverages and tobacco (~10%).

The consumption of goods and services from the “construction” sector was responsible for 4.7 kg CO₂-eq./cap/day, also having a high contribution of several sectors. The largest contribution came from the production of non-metallic mineral products used as building materials (~36%), followed by the production and supply of the electricity used along the supply chain (~19%) and by the construction industry itself (~17%).

The demand for electricity generated 3.1 kg CO₂-eq./cap/day. Nearly all these emissions (more than 96%) occurred during the production, collection and distribution of the electricity required by the households.

Table 3 also presents the sectors in which the GHG emissions occur. There are three sectors with relative contributions higher than 10%: “production, collection and distribution of electricity” (25% of the total emissions), “land transport and transport by pipelines” (24% of the total emissions), and “agriculture, hunting, forestry and fishing” (13% of the total emissions).

Table 3
GHG emissions of Aveiro household consumption (kg CO₂-eq./cap/day).

Economic sectors demanding	Economic sectors emitting										Total
	Production, collection and distribution of electricity	Land transport; transport via pipelines	Agriculture, hunting, forestry and fishing	Other non-metallic mineral products	Chemicals excluding pharmaceuticals	Construction	Coke, refined petroleum products and nuclear fuel	Mining and quarrying (energy)	Food products, beverages and tobacco	Other sector ^a	
Land transport; transport via pipelines	0.193	4.970	0.015	0.028	0.050	0.009	0.185	0.124	0.001	0.066	5.64
Food products, beverages and tobacco	0.682	0.402	2.555	0.204	0.238	0.016	0.100	0.079	0.490	0.273	5.04
Construction	0.843	0.303	0.102	1.631	0.280	0.785	0.184	0.138	0.003	0.396	4.67
Production, collection and distribution of electricity	2.942	0.027	0.003	0.007	0.012	0.002	0.039	0.086	0.000	0.014	3.13
Hotels and restaurants	0.427	0.109	0.454	0.094	0.072	0.007	0.042	0.036	0.055	0.234	1.53
Motor vehicles, trailers and semi-trailers	0.231	0.108	0.021	0.107	0.098	0.004	0.023	0.020	0.001	0.136	0.75
Textiles, textile products, leather and footwear	0.191	0.069	0.044	0.014	0.111	0.003	0.020	0.017	0.002	0.239	0.71
Pharmaceuticals	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.696	0.70
Other manufacturing; recycling (including furniture)	0.185	0.092	0.060	0.048	0.096	0.004	0.026	0.021	0.001	0.151	0.68
Manufacture of gas; distribution of gaseous fuels through mains	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.672	0.67
Other community, social and personal services	0.302	0.079	0.023	0.023	0.044	0.007	0.026	0.023	0.001	0.113	0.64
Air transport	0.022	0.006	0.003	0.003	0.005	0.001	0.016	0.011	0.000	0.194	0.26
Pulp, paper, paper products, printing and publishing	0.046	0.023	0.027	0.005	0.023	0.001	0.005	0.004	0.001	0.099	0.23
Health and social work	0.042	0.013	0.023	0.005	0.056	0.001	0.016	0.011	0.002	0.053	0.22
Collection, purification and distribution of water	0.096	0.008	0.003	0.012	0.015	0.003	0.007	0.006	0.000	0.020	0.17
Post and telecommunications	0.052	0.023	0.006	0.009	0.012	0.003	0.005	0.005	0.000	0.042	0.16
Other electrical machinery and apparatus	0.044	0.029	0.004	0.008	0.030	0.001	0.005	0.004	0.000	0.026	0.15
Wood and products of wood and cork	0.015	0.011	0.034	0.002	0.007	0.000	0.002	0.002	0.000	0.037	0.11
Radio, television and communication equipment	0.029	0.017	0.003	0.006	0.015	0.001	0.004	0.003	0.000	0.032	0.11
Other machinery and equipment	0.032	0.014	0.002	0.009	0.009	0.001	0.003	0.003	0.000	0.024	0.10
Finance and insurance	0.027	0.008	0.004	0.003	0.004	0.001	0.002	0.002	0.000	0.017	0.07
Education	0.022	0.004	0.002	0.001	0.002	0.000	0.002	0.001	0.000	0.011	0.05
TOTAL	6.42	6.31	3.39	2.22	1.18	0.85	0.71	0.60	0.56	3.55	25.79

^a Sum of all the other sectors of the IO table.

The total estimated fossil fuel consumption to satisfy household demands was 7.3 koe/cap/day (Table 4). The more contributing sectors to this consumption are the same mentioned above for GHG emissions, although the relative contributions of each sector differ. Thus, the consumption of goods and services from the sector of “land transport and transport by pipelines” contributed with 25% for this energy consumption, followed by the sectors of “construction” (18% of the total fossil fuel consumption), “food products, beverages and tobacco” (14% of the total fossil fuel consumption), and “production, collection and distribution of electricity” (12% of the total fossil fuel consumption). Regarding the activities that consume more fossil fuels within each sector, they are the same referred for GHG emissions, although the relative contributions are not exactly the same, because GHG emissions may arise from processes other than fossil fuel combustion. This is the reason why, contrastingly to GHG emissions, the sector of “agriculture, hunting, forestry and fishing” is not relevant for fossil fuel use. In fact, GHG emissions from this sector mainly consist in methane and nitrous oxide derived from enteric fermentation, manure management and agricultural soils (Maciel et al., 2012).

Regarding the sectors where fossil fuel consumption took place, the two most important were the same found for GHG emissions, because GHG emissions from these sectors mostly derive from fossil fuel combustion. These sectors are “land transport and transport by pipelines” (29% of the total fossil fuel consumption) and “production, collection and distribution of electricity” (24% of the total fossil fuel consumption).

The results of the sensitivity analysis show that the data used for the household expenditure can have a relatively significant effect on GHG emissions and fossil fuel consumption. The results obtained in Scenario A, in which average national expenditure data were directly adopted, amounted to 19 kg CO₂-eq./cap/day and 5.5 koe/cap/day, representing a decrease of 26% in relation to the base scenario in both cases, as a result of smaller household expenditures in Portugal than in Aveiro (Table 1). Despite these differences, the relative contribution of each economic sector to each impact varies only slightly in relation to the base scenario (<1%), as shown in Fig. 2 from the perspective of the most demanding sectors. This is explained by the fact that the relative contribution of each sector to the household expenditure in Aveiro is very similar to that in Portugal (Table 1).

Scenario B (that considers the average national expenditure multiplied by 1.34) leads to impacts that differ from those obtained in the base scenario by less than 0.5%. Finally, in Scenario C (that adopts expenditures disaggregated by age of the reference person but ignores that the purchase power is 1.34 times higher in Aveiro than the national purchase power), GHG emissions and fossil fuel consumption amounted to 20 kg CO₂-eq./cap/day and 5.7 koe/cap/day representing a decrease of 23 to 22% in relation to the base scenario, respectively. These results show that the effect of using a purchase power higher in Aveiro than the average purchase power in the country had a more significant effect in the final results than the effect of considering expenditures disaggregated by age of the reference person. Moreover, these results highlight the importance of using site-specific household expenditure data.

4. Discussion

This study employs environmentally extended IO analysis to determine GHG emissions and fossil fuel consumption associated with goods and services purchased by Aveiro households. For this purpose, a methodology adapted to the specificities of the data available is proposed. This section identifies and discusses the main limitations of this methodology, provides a comparison with other studies and proposes improvement measures to decrease the environmental impacts.

4.1. Limitations

The data required for developing the environmentally extended IO analysis (i.e. IO table, data on environmental impacts and data on household expenditure) were only available at the national scale. Ideally, household expenditure surveys should be available at a finer resolution level to allow downscaling of the national impacts to the urban level, but this was not the case. Therefore, a procedure was suggested to overcome this limitation, based on the consideration of three data sets: (1) national household expenditure data disaggregated per age of reference person, (2) number of families living in Aveiro per age of reference person and (3) ratio between the purchase power in Aveiro and Portugal.

The sensitivity analysis performed in this study shows that this downscaling procedure can lead to variations in the total GHG emissions and fossil fuel consumption of about 22–26%, thus introducing some uncertainty. Therefore, efforts should be undertaken in the future to elaborate household expenditure surveys specific for Aveiro in order to decrease uncertainty. This limitation is also recognised by other authors, such as Minx et al. (2009), who recommend the use of geo-demographically segmented expenditure data. These data are geographically coded and contain economic and social variables allowing the distinction of several lifestyle groups based not only in characteristics such as income or occupation, but also taking into account neighborhood and community characteristics such as the type and size of dwelling, access to services etc. Such distinction is very relevant in environmentally extended IO analysis since lifestyles influence expenditure patterns and consequently environmental impacts (Baicocchi et al., 2010; Druckman and Jackson, 2009; Lenzen et al., 2004a; Minx et al., 2009). Moreover, given the heterogeneity of lifestyles in Aveiro derived from the existence of urban-like and rural-like parish subdivisions, the application of this kind of data would produce results more relevant for policy making. Providing information on geographic and lifestyle based distribution of the impacts would allow the implementation of more effective improvement actions for specific target-groups and areas (Minx et al., 2009) but, at the same time, it would increase the complexity of the methodology.

In addition, expenditure data should be disaggregated by sectors in such a manner that easily allows the establishment of equivalencies with the sectors of the IO table. Currently, the sectors do not exactly match and the establishment of equivalences between sectors was needed (Table A.1), which may introduce some distortions in the allocation of impacts to the economic sectors. The most critical aspect is the assignment of all the expenditures related to buying and renting a house to the construction sector, which overestimates the impacts of this sector and, in a smaller extent (due to small environmental intensities), underestimates the impacts of the finance and insurance sector (paying rents, etc.). Only buying or renting a newly built house should be allocated to the construction sector, being the rest allocated to the finance and insurance sector. Therefore, household expenditures related with buying or renting houses should distinguish between money spent on newly built and old houses in order to reduce the uncertainty associated with the results obtained for the construction and the finance and insurance sectors.

The waste and wastewater management sector was excluded from the current study because it could not be assigned directly to any economic sector of the IO table. In Portugal, average direct emissions from this sector amounted to about 1.5 kg CO₂-eq./cap/day (INE, 2011a). Since waste and wastewater management practices in Aveiro did not greatly differ from the waste and wastewater management practices in the majority of the Portuguese municipalities, the inclusion of this sector would only slightly (~5%)

Table 4
Fossil fuel consumption of Aveiro household consumption (koe/cap/day).

Economic sectors demanding	Economic sectors emitting										Total
	Land transport; transport via pipelines	Production, collection and distribution of electricity	Other non-metallic mineral products	Pulp, paper, paper products, printing and publishing	Construction	Food products, beverages and tobacco	Chemicals excluding pharmaceuticals	Mining and quarrying (energy)	Coke, refined petroleum products and nuclear fuel	Other sector ^a	
Land transport; transport via pipelines	1.652	0.052	0.006	0.011	0.003	0.001	0.012	0.048	0.054	0.027	1.864
Construction	0.101	0.227	0.348	0.032	0.268	0.002	0.064	0.053	0.054	0.166	1.314
Food products, beverages and tobacco	0.134	0.183	0.044	0.089	0.006	0.239	0.055	0.031	0.029	0.196	1.003
Production, collection and distribution of electricity	0.009	0.791	0.002	0.003	0.001	0.000	0.003	0.033	0.011	0.006	0.858
Hotels and restaurants	0.036	0.115	0.020	0.024	0.002	0.027	0.016	0.014	0.012	0.159	0.426
Motor vehicles, trailers and semi-trailers	0.036	0.062	0.023	0.011	0.001	0.001	0.022	0.008	0.007	0.077	0.248
Textiles, textile products, leather and footwear	0.023	0.051	0.003	0.011	0.001	0.001	0.025	0.007	0.006	0.116	0.244
Other community, social and personal services	0.026	0.081	0.005	0.019	0.002	0.001	0.010	0.009	0.007	0.069	0.230
Other manufacturing; recycling (including furniture)	0.031	0.050	0.010	0.013	0.001	0.001	0.022	0.008	0.008	0.065	0.208
Manufacture of gas; distribution of gaseous fuels through mains	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.181	0.181
Pharmaceuticals	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.160	0.160
Pulp, paper, paper products, printing and publishing	0.008	0.012	0.001	0.098	0.000	0.000	0.005	0.002	0.001	0.005	0.133
Air transport	0.002	0.006	0.001	0.001	0.000	0.000	0.001	0.004	0.005	0.066	0.086
Health and social work	0.004	0.011	0.001	0.003	0.000	0.001	0.013	0.004	0.005	0.025	0.069
Post and telecommunications	0.008	0.014	0.002	0.008	0.001	0.000	0.003	0.002	0.002	0.024	0.063
Collection, purification and distribution of water	0.003	0.026	0.003	0.002	0.001	0.000	0.003	0.002	0.002	0.017	0.059
Other electrical machinery and apparatus	0.010	0.012	0.002	0.003	0.000	0.000	0.007	0.002	0.002	0.018	0.054
Radio, television and communication equipment	0.006	0.008	0.001	0.003	0.000	0.000	0.004	0.001	0.001	0.020	0.044
Other machinery and equipment	0.005	0.009	0.002	0.001	0.000	0.000	0.002	0.001	0.001	0.012	0.033
Finance and insurance	0.003	0.007	0.001	0.004	0.000	0.000	0.001	0.001	0.001	0.009	0.027
Wood and products of wood and cork	0.004	0.004	0.001	0.001	0.000	0.000	0.002	0.001	0.001	0.011	0.023
Education	0.001	0.006	0.000	0.002	0.000	0.000	0.000	0.001	0.000	0.006	0.017
Total	2.099	1.726	0.474	0.339	0.289	0.273	0.270	0.230	0.207	1.435	7.343

^a Sum of all the other sectors of the IO table.

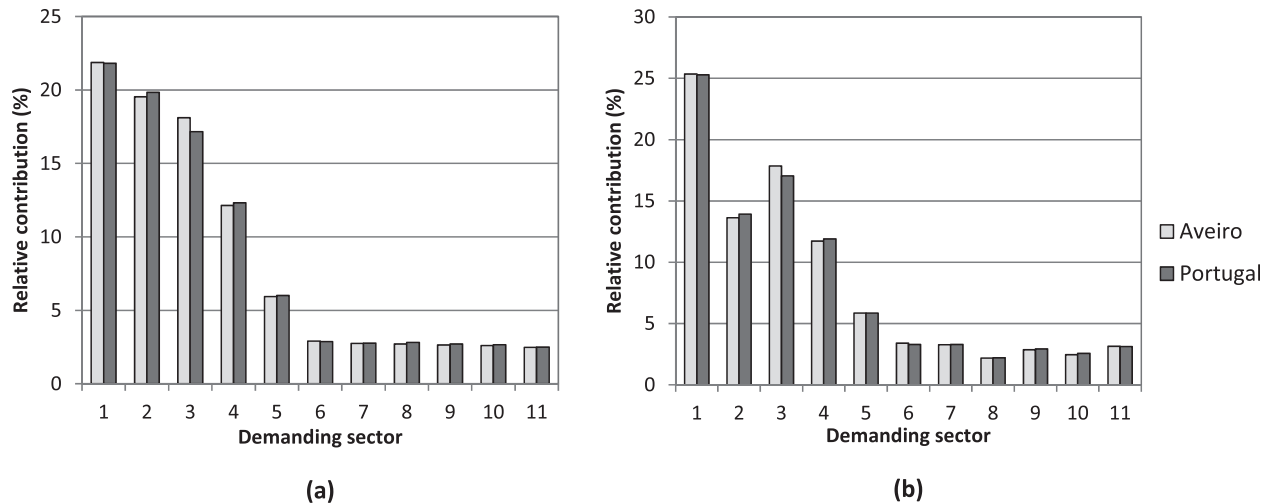


Fig. 2. Relative contribution of the most relevant demanding sectors for GHG emissions (a) and fossil fuel consumption (b). 1 = Land transport; transport via pipelines. 2 = Food products, beverages and tobacco. 3 = Construction. 4 = Production, collection and distribution of electricity. 5 = Hotels and restaurants. 6 = Motor vehicles, trailers and semi-trailers. 7 = Textiles, textile products, leather and footwear. 8 = Pharmaceuticals. 9 = Other manufacturing; recycling (including furniture). 10 = Manufacture of gas; distribution of gaseous fuels through mains. 11 = Other community, social and personal services.

increase total GHG emissions of Aveiro. Similarly, fossil fuel consumption of this sector would not likely be significant.

Temporal discrepancy of datasets is also a limitation of this study. All data refer to the year 2005 except data on the number of families in Aveiro used to derive municipal expenditures, which refers to the year 2001. During the period comprised between 2001 and 2011 (when population census were performed) the number of families in Aveiro increased almost 2% per year (INE, 2002; INE, 2012) and assuming linearity this means an increase of about 8% in the number of families between 2001 and 2005, which also adds some bias to the results obtained.

The assumption that the environmental intensities associated with imported goods is identical to those produced in Portugal, which is inherent to the use of a single-region IO table, can introduce errors, as shown by Lenzen et al. (2004b). This shortcoming can be overcome by employing multi-region IO tables and different impact intensities for imported goods (Davis and Caldeira, 2010; Lenzen et al., 2004b; Su and Ang, 2014; Wiebe et al., 2012; Wiedmann, 2009), but this framework also poses additional challenges related with data availability, data reconciliation and computability (Wiedmann et al., 2010).

Another source of uncertainty related with impact intensities is the assumption that economic sectors in the IO table without data on gross outputs and economic sectors without data on annual GHG emissions and fossil fuel consumption have the same impact intensities of similar sectors (Table A.2). However, the effect of this assumption is not relevant as none of these sectors with assumed impact intensities has contributions higher than 3% to the total results.

Other sources of uncertainty normally associated with environmentally extended IO analysis pointed out by other authors (e.g. Lenzen et al., 2004a; Rosado and Ferrão, 2009; Wiedmann, 2009) that are not specific of this study but also affect our results include sampling and reporting errors in basic source data, and the assumption of proportionality between monetary and physical flows.

4.2. Comparison with other studies

Total fossil fuel consumption and GHG emissions resulting from the private consumption of Aveiro dwellers amounted to 7.3 koe/cap/day and 26 kg CO₂-eq./cap/day, respectively, in 2005. The

results obtained for Portugal in the same year were smaller, 5.5 koe/cap/day and 19 kg CO₂-eq./cap/day. Two other studies based also on environmentally extended IO analysis, one performed for Portugal (Cruz and Barata, 2008) and the other for Lisbon, the Portuguese capital (Rosado and Ferrão, 2009), obtained some divergent results from ours. Cruz and Barata (2008) obtained a much lower emission of CO₂ (~10 kg CO₂/cap/day) associated with the consumption of goods and services by the total Portuguese households. However, estimations were undertaken for a different year (1999) and GHGs other than CO₂ were excluded. These GHGs should account for about 20–25% of the total GHGs (Maciel et al., 2012). Despite these differences, according to Cruz and Barata (2008) the sectors that contribute most to CO₂ emissions are the same identified in the current study: construction, electricity production and supply, transports, and manufacture of food and beverages.

Rosado and Ferrão (2009) estimated the energy use and CO₂ emissions triggered by the private consumption of households living in Lisbon. Their study was limited to some products such as food, beverages, textiles, wood, paper, chemicals, non-metallic mineral products, motor vehicles, etc. Within these sectors, the one with the highest energy use and CO₂ emissions was the sector of “food products, beverages and tobacco”, as in the present study, but with contributions that are about half (0.5 koe/cap/day) and one third (1.6 kg CO₂/cap/day) of those obtained for Aveiro for energy use and CO₂ emissions, respectively. Despite the likely differences in household consumption patterns in Aveiro and Lisbon, discrepancies in the estimates may also arise from the consideration of different years of analysis (2002 for Lisbon using the IO table of 1999 and 2005 for Aveiro) and methodological procedures (e.g. in the aggregation of sectors of the IO table, and downscaling of household expenditure data).

Using a bottom-up approach, based on activity data and GHG emission factors, Kennedy et al. (2009) found on their analysis of ten global cities, from a consumption perspective, GHG emissions ranging between 12 (Barcelona) and 59 (Denver) kg CO₂-eq./cap/day. The magnitude of these emissions depended on a number of factors such as climate, access to resources, power generation mix, urban design, personal income and waste management practices. With 26 kg CO₂-eq./cap/day, Aveiro is within this range and close to its average. However, the boundaries of the two studies are not exactly the same, as Kennedy et al. (2009) limited

the analysis to seven sectors (electricity, heating and industrial fuels, industrial processes, ground transportation, aviation, marine and waste).

4.3. Improvement measures

A variety of policies and actions leading to reductions of the environmental impacts of cities, mostly focused on the transportation and energy sectors, have been proposed and analyzed in several studies (Hoornweg et al., 2011; Phdungsilp, 2010; World Bank, 2010). In Aveiro, we found that the sectors with the largest contributions to GHG emissions and fossil fuel consumption were those associated with transportation, consumption of electricity, construction and consumption of food products. Therefore, improvement measures should preferentially be established for these sectors in order to decrease more effectively the environmental impacts resulting from the demands of the households. Some measures depend directly on the household willingness, while others are out of their control and depend rather on existing urban infrastructure and policies, which influence lifestyle choices. For example, a lack of efficient public transportation system and low parking prices encourage greater car use (Hoornweg et al., 2011). Moreover, indirect impacts depend of national policies (e.g. electricity production mix) or corporate policies and performance. Examples of possible improvement measures targeted for the particular case of Aveiro are presented in Table 5 and discussed below.

Regarding the transportation sector, for which GHG emissions and fossil fuel consumption arise mostly during vehicle use, household choices may lead to important reductions of fossil fuel consumption and GHG emissions by choosing vehicles with lower fuel consumption, electric or hybrid, as well as by preferring public transport or non-motorised vehicles (bikes). Public transport has a great potential for increase as only about 10% of the population living in Aveiro uses public transport (INE, 2002). However, local authorities have also a relevant role in discouraging private vehicle use by improving public transport quality and by creating of safer conditions for bike circulation. Local

policies can also influence private vehicle choice for example by creating special parking privileges for less energy consuming and polluting vehicles.

Moreover, other studies suggest that GHG emissions and energy consumption are inversely correlated with urban population density (Kennedy et al., 2009; Parshall et al., 2010; Sugar et al., 2012). Hence, as Aveiro's population increases, GHG emissions and fossil fuel consumption per capita are expected to decrease. However, the relation between land use and travel behavior is complex, and other factors related with urban structure such as physical proximity to activities, jobs/housing balance, land use mix may also affect GHG emissions and fossil fuel consumption (Rickwood et al., 2008).

Most of the fossil fuel consumption and GHG emissions from electricity consumption are associated with electricity production in power plants. Although households have an important role in decreasing these impacts by reducing electricity consumption, GHG emissions from electricity consumption depend also on the GHG emission factors of the supply mix (Kennedy et al., 2009), which is beyond the control of the households. The electricity consumed in Aveiro is produced in the national grid and the supply mix is defined by national policies. In 2005, about 70% of the electricity consumed in the Portuguese grid was produced from fossil fuels. More specifically, 29% of the electricity was produced from hard coal, 25% from natural gas, 16% from fuel oil (DGEG, 2013). The remaining electricity was produced based on renewable sources (e.g. hydropower, wind power, biomass) and nuclear power (electricity imported). The corresponding GHG emission factor was about 390 g CO₂-eq./kWh, estimated as the quotient of direct GHG emissions from power plants (INE, 2011a) and total electricity produced in the national grid available for consumption (DGEG, 2013). An electricity mix with a lower percentage of fossil fuels would decrease both GHG emissions and fossil fuel consumption. This measure is already being implemented and the fossil fuel component of the Portuguese electricity mix has been changing from 70% in 2005 to 50% in 2011 (DGEG, 2013) and it is expected to decrease even more to 42% in 2020 (Seixas et al., 2012).

In the construction sector, the manufacture of building products presents a significant contribution to fossil fuel consumption and GHG emissions. Households and technicians (architects, engineers) play an important role in decreasing these impacts by choosing more sustainable building materials, preferably available locally (González and Navarro, 2006; Morel et al., 2001). Communication tools such as Environmental Product Declarations (e.g. from DAPHabitat, 2013) verified by independent entities should be promoted to facilitate the selection of building materials. Other measures to decrease these impacts should include national and corporate policies or strategies for setting building codes requiring more sustainable building materials and for promoting the use of best available techniques and eco-innovation in material manufacturing (Bribián et al., 2011; Hoornweg et al., 2011).

In the food sector, actions that could be implemented directly by the households to decrease their impacts include a more responsible consumption by decreasing food wastage and also the adoption of the Portuguese traditional Mediterranean-type diet (De Boer et al., 2006; Duchin, 2005). This diet is based on high consumption of olive oil, vegetables, fruits and cereals, moderately high consumption of fish and low consumption of meat and dairy products. Moreover, product labeling (e.g. carbon footprint) could support household decision in the selection of more sustainable products, although understanding of these labels by consumers is still low (Gadema and Oglethorpe, 2011; Vancly et al., 2011). Given that the food industry and the

Table 5
Improvement measures for decreasing GHG emissions and fossil fuel consumption in Aveiro.

Sector	Household decision	Policy tools
Transportation	- Purchase of more efficient, alternative fuel, electric or hybrid private vehicles - Use of public transportation - Use of non-motorised means of travel	- Creating special parking privileges for alternative fuel, electric or hybrid vehicles - Improve quality of public transportation - Increase bike lines
Electricity	- Decrease electricity consumption	- Moving to more renewable electricity mix
Construction	- Selection of more sustainable building materials	- Building codes requiring more sustainable building materials - Increase of industrial environmental performance (legislation, industry voluntary actions, etc.)
Food	- Decrease food wastage - Change diet	- Product labeling - Increase of industrial environmental performance (legislation, industry voluntary actions, etc.)

sector of agriculture, hunting, forestry and fishing have important shares in the fossil fuel consumption and GHG emissions of the food sector, improvement of current practices in these activities would also contribute to decrease the overall impact of food consumption. National and corporate policies should be implemented for this purpose (e.g., legislation, adoption of best practices and cleaner technologies).

5. Conclusion

Measuring the environmental impacts of cities is crucial for defining strategies leading to more sustainable cities. Environmentally extended IO analysis is well suited to assess the environmental impacts associated with consumption activities in cities allowing for a rapid assessment of a wide range of activities including upstream (indirect) impacts and identifying the sectors causing those impacts.

This study estimates direct and indirect GHG emissions and fossil fuel consumption derived from the private consumption of households living in the municipality of Aveiro, using environmentally extended IO analysis. As all data needed to apply this methodology are only available at the national scale, a downscaling procedure is proposed to obtain municipal GHG emissions and fossil fuel consumption. This procedure could also be applied to other municipalities or cities for which data on local household expenditures are missing.

The total GHG emissions estimated were 26 kg CO₂-eq./cap/day and the total fossil fuel consumption was 7.3 koe/cap/day. Four sectors played a major role in these environmental impacts: “land transport and transport by pipelines”, “food, beverages and tobacco”, “construction” and “production, collection and distribution of electricity”. This means that these are the sectors for which

policies should be primarily set in order to effectively decrease these impacts.

Further research work should try to enlarge the scope and improve the accuracy of the results obtained. This could be achieved by: (a) including public expenses (only household expenses were taken into account) and activities related with wastewater and solid waste management by using a Portuguese IO table disaggregated in more economic sectors, (b) using city scale or even lower scale information on private households expenditure, (c) attempting to use multi-region IO tables and country-specific impact intensities for imported goods. Several shortcomings of the method used in this study were detected and discussed. Therefore, another interesting topic for future research would be to apply other methods such as process LCA to estimate the same impacts in the municipality and to compare results, advantages and limitations of each method.

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Appendix

Table A.1

Equivalences between the 22 economic sectors of the IO table and the 47 sub-sectors of household expenditures database.

Economic sector of IO table	Sub-sector of household expenditures database
Construction	“Renting (monthly costs)”, “Renting (average costs estimated)” and “House maintenance”
Food products, beverages and tobacco	“Food products”, “Non-alcoholic drinks”, “Alcoholic drinks” and “Tobacco”
Hotels and restaurants	“Restaurants and similar” and “Hotels and similar”
Land transport; transport via pipelines	“Personal vehicles utilization costs” and “Transportation services”
Other community, social and personal services	“Cultural and recreation services”, “Hygiene and personal care services”, “Personal articles”, “Social protection services” and “Other services”
Motor vehicles, trailers and semi-trailers	“Acquisition of vehicles”
Textiles, textile products, leather and footwear	“Clothing” and “Footwear”
Other manufacturing; recycling (including furniture)	“Furniture, decoration and carpets”, “Domestic textile articles”, “Glass, dishes and domestic utilities”, “Home and gardening machinery and tools” and “Usual maintenance goods”
Pharmaceuticals	“Pharmaceuticals”
Post and telecommunications	“Postal services”, “Telecommunication” and “Phone services”
Production, collection and distribution of electricity	“Electricity, gas and other fuels” (65% of sub-sector total)
Health and social work	“Health services (except hospitals)” and “Hospitals”
Finance and insurance	“Insurance” and “Financial services”
Collection, purification and distribution of water	“Other services related with housing”
Education	“Basic school”, “High school”, “Technical school”, “University” and “Other education services”
Manufacture of gas; distribution of gaseous fuels through mains	“Electricity, gas and other fuels” (26% of sub-sector total)
Pulp, paper, paper products, printing and publishing	“Books, newspapers and other printed paper”
Radio, television and communication equipment	“Other recreational and leisure products and equipments”
Other electrical machinery and apparatus	“Audiovisual equipment, photographic and computer accessories”
Other machinery and equipment	“Domestic equipment”
Air transport	“Touristic travel”
Wood and products of wood and cork	“Electricity, gas and other fuels” (9% of sub-sector total)

Table A.2

Equivalences between the economic sectors of the IO table without data about GHG emission and fossil fuel intensities and the similar sectors used to fill the lack of these data

Economic sector without data about GHG emission and fossil fuel intensities	Similar economic sector used
Mining and quarrying (energy)	Mining and quarrying (non-energy)
Pharmaceuticals	Chemicals excluding pharmaceuticals
Non-ferrous metals	Iron and steel
Radio, television and communication equipment	Other machinery and equipment
Medical, precision and optical instruments	Other machinery and equipment
Aircraft and spacecraft	Building and repairing of ships and boats
Railroad equipment and transport equipment not elsewhere specified	Building and repairing of ships and boats
Manufacture of gas, distribution of gaseous fuels through mains	Production, collection and distribution of electricity

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